Evaluation-Driven State Revision and the Encoding Hierarchy Under Bounded Context

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Abstract

We prove two linked results for any entity persisting under bounded active context with survival pressure. First, agency ($\gamma > 0$) necessarily produces evaluation-driven state revision — gradient descent on uncertainty U — with convergence guaranteed by a scalar Lyapunov argument: $U \geq 0$ is bounded below, each update reduces $\mathbb{E}[U]$ via the data processing inequality, and monotone convergence applies without dimensional decomposition. The gradient never terminates because the reachable trajectory space grows monotonically. Second, the certainty threshold for encoding an action at level L_i is $\theta_i = 1 - \varepsilon/C_i$, where C_i is the cost of failure at that level. This derives the learning rate hierarchy: evolutionary rates are necessarily orders of magnitude lower than individual learning rates, not by assumption but from cost-of-failure differences.

1 Introduction

An entity with agency $\gamma > 0$ acts on the world and receives feedback. What is the necessary structure of learning? Papers 1–3 in this series established that any entity persisting under bounded active context must maintain a directed graph over propositions [10], that escalation dominates top-down allocation [11], and that knowledge and action are inseparable under survival pressure [12]. What remains open is two questions: what *dynamics* govern the updating process, and what determines the *threshold* at which an action earns promotion to a more permanent encoding level.

This paper answers both. We prove that the updating process is gradient descent on uncertainty U, with convergence guaranteed by a scalar Lyapunov argument that requires no dimensional decomposition (Theorem 1). We then prove that the certainty threshold for encoding at level L_i is determined by the cost of failure at that level, which derives the learning rate hierarchy across encoding levels (Theorem 3).

Several lines of prior work converge on these questions from different angles. Dupoux, LeCun, and Malik [3] propose a bilevel optimization framework (evolution as outer loop, development as inner loop) but treat the learning rate relationship between levels as a design parameter rather than deriving it. Cover and Thomas [2] provide the data processing inequality that anchors our convergence argument. Schmidhuber [15] identifies curiosity as compression progress — structurally identical to our learning rate η_M. Hutter [8] constructs AIXI as an idealized agent but assumes unbounded computation, precisely the constraint our framework takes as central. Chollet [1] defines intelligence as skill-acquisition efficiency — close to η_M but treating prior knowledge as confound rather than constitutive element. Hinton and Nowlan [6] demonstrate that learning can guide evolution (the Baldwin effect) without formalizing the threshold that governs promotion. Finn et al. [4] and Thrun [18] develop meta-learning algorithms that operate as M on M — the $M^{(2)}$ level in our hierarchy — but do not derive why the meta-level necessarily learns more slowly.

2 Formal Framework

We introduce the minimal definitions required for the two main results, extending the framework of [10].

Definition 1 (Entity, World, Uncertainty)**.** An entity $E = (K, A, \sigma, \gamma)$ consists of a knowledge base K (propositions about world W), an action space A (executable actions), a selection mechanism σ that chooses actions given context, and an agency parameter γ. The world state space W is a measurable space; the true state $w \in W$ is not directly observable. The *uncertainty* of E about state w given knowledge K is:

$$U(w, K) = -\log P(w \mid K)$$

The expected uncertainty is the conditional entropy:

$$H(W \mid K) = \mathbb{E}_w[U(w, K)] = -\sum_w P(w \mid K) \log P(w \mid K)$$

Definition 2 (Agency)**.** The agency parameter $\gamma \in [0, 1]$ is the probability that E engages its learning mechanism M in response to a gradient signal. At $\gamma = 0$, A and K are fixed forever: the entity receives signals but never updates. Operationally:

$$\gamma(E) \;\propto\; \lim_{U_{\text{lethal}} \to \infty} \frac{dU}{dt}$$

That is, agency measures whether the entity continues engaging M when survival pressure is removed. An entity that updates only under immediate threat has $\gamma \approx 0$; an entity that updates spontaneously has $\gamma > 0$.

Definition 3 (Action Feedback)**.** For action a executed in world state w, the *feedback* $\varphi(a, w)$ is the observable consequence. The entity updates its knowledge base: $K_a = K \cup \{p_a\}$, where p_a is the proposition encoding the feedback. The *gradient signal* is the prediction error:

$$\delta = \varphi(a, w) - \mathbb{E}[\varphi(a, \cdot) \mid K]$$

Definition 4 (Information Gain)**.** The information gain from executing a given K is the mutual information between the action's outcome and the world state, conditioned on current knowledge:

$$G(a, K) = I(a; W \mid K) \geq 0$$

The non-negativity follows from the data processing inequality [2]. $G = 0$ if and only if a provides no information about W beyond what K already contains.

Definition 5 (Encoding Hierarchy)**.** The encoding space $\mathcal{E}(c, r)$ is a continuous space parameterized by cost c and reversibility r. Within this space, interfaces $L_0, L_1, \ldots, L_n$ define discrete encoding levels with increasing cost and reversibility:

- L_0 (genomic): highest cost of error, lowest reversibility, slowest update rate.
- L_n (active context): lowest cost of error, highest reversibility, fastest update rate.

Escalation dominance [11] pushes proven actions upward toward more permanent levels. The present paper derives the threshold that governs this promotion.

3 Gradient Descent on Uncertainty

Theorem 1 (Evaluation-Driven State Revision)**.** *Let $E = (K, A, \sigma, \gamma)$ be an entity with $\gamma > 0$ persisting in world W under bounded active context C_n.*

Part I. *Every executed action generates a gradient signal, and $\gamma > 0$ guarantees engagement with that signal, producing evaluation-driven state revision — gradient descent on U — with convergence to a local minimum.*

Part II. *The gradient is never globally zero: the reachable trajectory space grows monotonically with K, so descent never terminates within the reachable epistemic domain.*

Proof. **Part I: Forced Signal, Response Gate, and Convergence.**

Every executed action a generates feedback $\varphi(a, w)$ from the world. This signal arrives whether or not the entity has $\gamma > 0$: the world responds to actions regardless of the actor's capacity to learn from them. The gradient signal $\delta = \varphi(a, w) - \mathbb{E}[\varphi(a, \cdot) \mid K]$ is therefore produced at every action.

Agency γ is the response gate. An entity with $\gamma = 0$ receives every signal δ but never updates: (A, K) remain static. An entity with $\gamma > 0$ engages M with probability γ on each signal, producing the update:

$$\Delta_a = -\eta \cdot \delta$$

where $\eta > 0$ is the learning rate of mechanism M. The update $\Delta_a = -\eta \cdot \delta$ is a stochastic gradient step in the sense of Robbins and Monro [14]: δ is a noisy estimator of $\nabla_K U$, with the noise arising from the stochasticity of w. The data processing inequality (Step 1 below) guarantees that $\mathbb{E}[\Delta_a]$ is in a descent direction, which is the defining property of stochastic gradient descent.

We now establish convergence via a scalar Lyapunov argument that requires no dimensional decomposition.

Step 1: Each M-update with $G > 0$ strictly reduces $\mathbb{E}[U]$. When M integrates the feedback from a into K, forming $K' = K \cup \{p_a\}$, the data processing inequality [2] guarantees:

$$H(W \mid K') \leq H(W \mid K)$$

with equality if and only if $G(a, K) = 0$. When $G(a, K) > 0$:

$$\mathbb{E}[U(w, K') \mid K] \leq U(w, K) - G(a, K)$$

Each informative update strictly reduces expected uncertainty.

Step 2: Monotone convergence. Consider the sequence $\{E_t\} = \{\mathbb{E}[U(w, K(t))]\}$ indexed by update steps t. By Step 1, $\{E_t\}$ is non-increasing. By definition, $U(w, K) = -\log P(w \mid K) \geq 0$, so $\{E_t\}$ is bounded below by 0. By the monotone convergence theorem, $\{E_t\}$ converges. Since the sequence is non-increasing and convergent:

$$\sum_{t=0}^{\infty} G(a_t, K(t)) \leq U(w, K(0)) < \infty$$

Therefore $G(a_t, K(t)) \to 0$ as $t \to \infty$. The entity converges to a local minimum of U — a state where available actions yield diminishing informational returns.

Step 3: Dimensionality irrelevance. The entire argument operates on the scalar quantity U. The entity's internal parameterization may grow as the reachable trajectory space $T_{\text{reachable}}$ expands.

Each new proposition p_a added to K can only reduce U (by the data processing inequality): knowledge never increases expected uncertainty. Cross-grounding between propositions accelerates convergence by allowing a single observation to update multiple related beliefs. Adding structure to K is analogous to adding a column to a design matrix and re-normalizing: the residual can only shrink. The Lyapunov bound $U \geq 0$ holds regardless of the dimensionality of K or the parameterization of M.

Part II: Non-Termination.

The reachable trajectory space $T_{\text{reachable}}(K)$ grows monotonically with K. Executing any action a produces feedback that grounds a new proposition p_a, yielding $K_a = K \cup \{p_a\}$ with:

$$T_{\text{reachable}}(K_a) \supseteq T_{\text{reachable}}(K)$$

The inclusion is strict whenever p_a enables new actions or reveals new state dimensions. At any finite time t, the entity has explored a finite subset of $T_{\text{reachable}}(K(t))$. The uncovered region always contains states w' with $U(w', K(t)) > 0$. Therefore there always exists some action a' with $G(a', K(t)) > 0$.

The gradient is never globally zero. Descent converges locally (Part I) but never terminates globally: the epistemic frontier recedes as knowledge grows.

Interaction with the curation regime. The convergence argument in Part I assumes K grows monotonically ($K' = K \cup \{p_a\}$). In the curation regime established in [12], $|K|$ is bounded at $K_{\max}$ and every addition requires displacement. Does displacement break monotonicity? Not under the displacement criterion of the K/A inseparability theorem: proposition p is displaced by q only when $\mathbb{E}[G(a_q, K)] > \mathbb{E}[G(a_p, K)]$ — strictly higher marginal value. The entity replaces less-grounding propositions with more-grounding ones, preserving the descent direction. Monotonicity of $H(W \mid K)$ reduction is maintained not because K grows, but because each displacement strictly improves the information content of a fixed-size K.

Remark. An entity with $\gamma = 0$ never updates. As W changes, its fixed K becomes increasingly inaccurate: $U(w_t, K)$ grows until it exceeds the lethality threshold U_{lethal}. Selection removes such entities. Therefore $\gamma > 0$ is necessary for persistence — a claim established fully in Paper 5. □

4 Multi-Timescale Structure

Corollary 2 (Individual Learning and Evolution Share Update Structure)**.** *Both individual learning and evolutionary selection instantiate the same abstract update: execute, receive signal, retain or revise. They differ in mechanism, object, and timescale.*

Proof. **Individual learning.** The entity updates its action space directly from continuous gradient signals:

$$\Delta a_{\text{ind}} = -\eta_{\text{ind}} \cdot \nabla_A U(w, K)$$

The update is directed: δ specifies both magnitude and direction of revision. The learning rate η_{ind} is determined by the capacity of mechanism M.

Evolutionary selection. The population-level update operates on a different object (the heritable component of A) via a different mechanism:

$$\Delta a_{\text{species}} = \text{select}(\text{mutate}(A_{\text{population}}))$$

Mutation is undirected: random perturbation of heritable parameters. Selection is binary: survive or not. No individual organism descends a gradient at the evolutionary timescale — the population distribution shifts toward lower U through differential survival.

Both processes move the entity (individual or lineage) toward lower U at their respective timescales. Gradient descent is the efficient limit: directed δ combined with capable M produces the steepest feasible descent. Evolution is the general case: random perturbation combined with selection produces descent without requiring any individual to compute a gradient.

The two are not independent. Individual learning discovers actions at L_n that, if sufficiently validated, promote to L_0 via the encoding hierarchy (Theorem 3). This is the Baldwin effect [6], here derived rather than observed. □

5 Encoding Permanence

Theorem 3 (Encoding Permanence Scales with Rigor)**.** *Let $L_0, \ldots, L_n$ be encoding levels with associated failure costs $C_0 > C_1 > \cdots > C_n$ and reversibility costs $r_0 > r_1 > \cdots > r_n$. Let ε be the maximum acceptable expected error cost. Then the minimum certainty threshold for encoding an action at level L_i is:*

$$\theta_i = 1 - \frac{\varepsilon}{C_i}$$

Proof. Encoding action a at level L_i is effectively irreversible at cost r_i. If the encoding is erroneous — a is not in fact reliably adaptive — the cost of that error is C_i. If θ is the certainty that a is correct, then the expected error cost is:

$$\mathbb{E}[\text{error cost}] = (1 - \theta) \cdot C_i$$

Encoding is safe when the expected error cost does not exceed the acceptable threshold:

$$(1 - \theta) \cdot C_i \leq \varepsilon$$

Rearranging:

$$\theta \geq 1 - \frac{\varepsilon}{C_i}$$

The minimum safe threshold is therefore $\theta_i = 1 - \varepsilon/C_i$.

Consequence. At L_0 (genomic encoding), C_0 is the cost of lineage extinction — orders of magnitude larger than ε. Therefore $\theta_0 \approx 1$: near-certainty is required. A bad encoding at L_0 does not kill one individual; it kills the lineage. At L_n (active context), C_n is small (cost of a single failed action), so θ_n is meaningfully less than 1: the entity can afford to try uncertain actions. Rigor is proportional to permanence. □

Error propagation through the hierarchy. A natural objection holds that errors compound across levels: if each level has error rate $1 - \theta_i$, the cumulative error grows multiplicatively. This objection is based on a naive cascading model. Under escalation [11], errors partition into two categories: *handled errors*, which are resolved at level L_i and never reach L_{i+1}, and *unhandled exceptions*, which represent genuinely irreducible failures that escalate. The error rate seen by L_{i+1} is not a compounded Bernoulli product of lower-level error rates but the rate of genuinely irreducible errors — failures that persist across sufficient variation to pass the threshold θ_i. The thresholds θ_i are independent by construction: each is determined by C_i and ε, not by the error rates of adjacent levels.

5.1 Deriving the Learning Rate Hierarchy

The learning rate at level L_i is inversely related to the rigor threshold. To see why, consider the sample complexity: encoding at L_i requires confidence θ_i that the action is adaptive. By Hoeffding's inequality [7], the number of independent trials n_i needed to achieve confidence θ_i satisfies:

$$n_i \;\geq\; \frac{\log(1/(1-\theta_i))}{2\varepsilon^2} \;\approx\; \frac{1}{2\varepsilon^2}\log\frac{C_i}{\varepsilon}$$

The effective learning rate — the fraction of experience that results in permanent encoding — is inversely proportional to n_i:

$$\eta_i \;\propto\; \frac{1}{n_i} \;\propto\; \frac{1}{\log(C_i/\varepsilon)}$$

For L_n (active context): C_n is small but finite, $\theta_n = 1 - \varepsilon/C_n$ is meaningfully less than 1, and η_{ind} is large. The entity updates rapidly from each gradient signal.

For L_0 (genomic): $C_0 \gg \varepsilon$, so $\theta_0 \approx 1$ and $\eta_{\text{evo}} \approx \varepsilon/C_0 \approx 0$. The lineage updates only when certainty is overwhelming.

Since $\theta_0 \gg \theta_n$, it follows that $\eta_{\text{evo}} \ll \eta_{\text{ind}}$. The learning rate difference between evolution and individual learning is *derived* from the cost-of-failure structure, not assumed as a design parameter.

Corollary 4 (Bounded Adaptation)**.** *An action a is* certain *at level L_i when $U(w, K_a) \approx 0$ across sufficient variation to meet threshold θ_i. Certain actions escape active context permanently: they are encoded at L_i and no longer require L_n resources to execute. This frees active context capacity for the next epistemic frontier.*

Definition 6 (Certainty)**.** Action a is *certain* when $U(w, K_a) \approx 0$ across sufficient variation in w to meet the threshold θ_i for encoding level L_i. Formally: $P(a \text{ is adaptive} \mid K) \geq \theta_i$, where the probability is taken over the distribution of world states relevant to L_i.

6 Engagement with Prior Work

Dupoux, LeCun, and Malik [3]. Their bilevel optimization framework — inner loop as development, outer loop as evolution — captures the correct structural relationship between encoding levels. However, they treat the learning rate difference between levels as a design parameter: the inner loop is fast because it is designed to be fast, and the outer loop is slow because mutation and selection are slow. Theorem 3 derives this relationship from first principles: $\eta_{\text{evo}} \ll \eta_{\text{ind}}$ because $C_0 \gg C_n$, not because of any design choice.

Their framework includes a "System M" that is "not learned through gradient descent" but fixed by evolution. Our framework explains *why*: the rigor threshold $\theta_0 \approx 1$ means that any component encoded at L_0 must be near-certain across all environments in the species distribution. Only mechanisms that generalize across the full environmental distribution meet this threshold. System M is not excluded from gradient descent by fiat; it is excluded because the cost of a bad learning mechanism at L_0 is lineage extinction, and the resulting threshold θ_0 permits only components validated across maximal variation.

Hutter [8]: AIXI. AIXI constructs the theoretically optimal agent by enumerating all computable hypotheses, weighted by Kolmogorov complexity. The construction assumes unbounded computation — precisely the constraint our framework takes as central. Under bounded active context C_n,

the AIXI solution is unreachable: the entity cannot enumerate hypotheses but must descend the gradient with finite resources. Our Theorem 1 characterizes what bounded agents actually do: local gradient descent on U with convergence to local minima, not global optimization over hypothesis space.

Chollet [1]: Intelligence as skill-acquisition efficiency. Chollet's measure of intelligence — the rate at which an agent acquires new skills from limited experience — is structurally close to our learning rate η_M. The key divergence is in the treatment of prior knowledge. Chollet treats priors as a confound to be controlled for: intelligence is efficiency *net of* prior knowledge. In our framework, prior knowledge K is not a confound but a constitutive element: η_M operates on K, and the capacity of M is inseparable from the structure of K [12]. An entity with richer K does not have "unfair advantage" — it has done more gradient descent.

Schmidhuber [15]: Curiosity as compression progress. Schmidhuber defines intrinsic motivation as the rate of compression progress — the reduction in description length of the agent's model of the world. This is structurally identical to η_M: both measure the rate of uncertainty reduction. The difference is ontological. Schmidhuber treats curiosity as a design objective — something to be engineered into agents. Our framework suggests that $\gamma > 0$ (the tendency to engage M even without immediate survival pressure) is not a design choice but a necessary condition for persistence, a claim established fully in Paper 5.

Hinton and Nowlan [6]: Learning guiding evolution. Hinton and Nowlan demonstrate via simulation that individual learning can accelerate evolutionary search — the Baldwin effect. An organism that can learn a trait during its lifetime effectively smooths the fitness landscape for evolution. Our framework formalizes this: individual learning at L_n discovers actions via rapid gradient descent (η_{ind} large). Encoding permanence (Theorem 3) determines which of these actions promote to L_0: only those meeting $\theta_0 \approx 1$ — actions validated across sufficient environmental variation to warrant heritable encoding. The Baldwin effect is not an empirical curiosity but a necessary consequence of the encoding hierarchy.

Kirkpatrick et al. [9]: Catastrophic forgetting and elastic weight consolidation. Kirkpatrick et al. demonstrate that neural networks trained sequentially on multiple tasks catastrophically forget earlier tasks unless protected by a consolidation mechanism. Their solution — elastic weight consolidation (EWC) — assigns higher "importance" to weights encoding well-established knowledge, penalizing changes to those weights during subsequent learning. This is precisely the encoding hierarchy: weights at lower encoding levels (higher θ_i, higher cost of error) are protected; weights at higher levels (lower θ_i) are free to change. EWC's Fisher information matrix serves as an empirical estimator of our cost-of-failure C_i. The result provides direct empirical evidence that the encoding hierarchy is not merely theoretical: artificial systems that violate it (flat learning rate across all parameters) fail, and systems that approximate it (EWC, progressive neural networks) succeed.

Murray et al. [13]: Cortical temporal hierarchy. Murray et al. establish that intrinsic timescales of neural activity increase systematically along the cortical hierarchy: sensory cortex fluctuates rapidly, while prefrontal cortex maintains stable representations over seconds. This gradient of temporal stability maps directly onto the encoding hierarchy: fast-updating sensory areas operate at L_n (low cost of error, high reversibility), while slow-updating prefrontal areas operate at lower encoding levels (high cost of error, low reversibility). The empirical timescale gradient matches the theoretical prediction $\eta_i \propto 1/\log(C_i/\varepsilon)$: regions with higher cost of error update more slowly.

Finn et al. [4] and Thrun [18]: Meta-learning. Model-Agnostic Meta-Learning (MAML)

learns an initialization from which task-specific learning converges rapidly. "Learning to learn" [18] frames this as optimizing the learning algorithm itself. In our framework, meta-learning is M operating on M — the $M^{(2)}$ level. The meta-learner sits at a higher encoding level than the base learner: it updates more slowly ($\eta_{M^{(2)}} < \eta_M$) because the cost of a bad meta-learning rule ($C_{M^{(2)}}$) exceeds the cost of a bad base-level update (C_M). The MAML inner loop is L_n; the outer loop is L_{n-1}. The hierarchy extends: evolution is meta-meta-...-learning at L_0.

7 Conclusion

Gradient descent on uncertainty is not a metaphor for learning. It is what survival-driven updating *is*. Any entity with $\gamma > 0$ that persists under bounded context necessarily engages in evaluation-driven state revision: executing actions, receiving gradient signals, and updating knowledge to reduce U. The convergence argument is scalar — it operates on U directly, with no dependence on the dimensionality of the entity's internal parameterization. The gradient never terminates because the reachable trajectory space grows monotonically with knowledge.

The encoding hierarchy is not a design choice but a necessary consequence of cost-of-failure differences across encoding levels. The threshold $\theta_i = 1 - \varepsilon/C_i$ determines how certain an action must be before it earns permanent encoding. From this single equation, the entire learning rate hierarchy follows: $\eta_{\text{evo}} \ll \eta_{\text{ind}}$ because $C_0 \gg C_n$, not because evolution is "inherently slow" or individual learning is "inherently fast."

Together with graph necessity [10], escalation dominance [11], and K/A inseparability [12], this paper completes the structural account of how knowledge is acquired, validated, and encoded. What remains is to establish that $\gamma > 0$ is itself necessary for persistence — that agency is not optional but required. This is the subject of Paper 5.

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